

Perceived Income Risk and Its Dynamics Across the Income Distribution: Persistence, Asymmetry, and a Diagnostic Aiyagari Model

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Heterogeneous-agent models need an income risk process. Standard practice: pin σ^2 down from realized PSID and hold it fixed.

Two assumptions hidden in that step:

- 1 Perceived risk equals objective risk
- 2 Perceived risk is constant: in level, in dynamics, and in response to news

This paper. Uses NY Fed SCE density forecasts to measure perceived σ^2 in a panel and characterize its *dynamics*. How persistent is it? How does it respond to surprises? Does it differ across the income distribution?

Why it matters. Aggregate consequence: perceived risk stays elevated past the income recovery, so precautionary saving drags the consumption rebound, especially at the bottom where MPCs are highest. Counter-point to Bellifemine, Couturier & Hosseini (2025) overoptimism, a positive shock can still raise saving here too.

Three empirical findings

Three facts about perceived income risk across the income distribution:

- ① **Levels are income-graded.** Bottom perceives $\sim 2.5\times$ the variance of Top.
- ② **Persistence is income-graded.** Bottom carries last quarter's risk forward $\sim 3\times$ as strongly as Top.
- ③ **Asymmetry concentrates at the Bottom.** Bottom updates $\sim 2.4\times$ more on bad news than on equal good news. Middle and Top symmetric.

⇒ **Not in realized income.** PSID income process flat across groups; the gradient lives entirely in beliefs.

⇒ **One mechanism delivers all three.** Buffer-stock with state-dependent attention $\phi(a, y)$ and asymmetric weight on bad news. Cross-tercile gradient is a composition effect of the joint (a, y) distribution, not tercile-conditional primitives.

Empirical Facts

The density forecast question

Suppose again that, **12 months from now**, you are working in the exact same ["main" if Q11>1] job at the same place you currently work, and working the exact same number of hours. In your view, what would you say is the percent chance that 12 months from now...

Instruction H4.

Your earnings on this job, before taxes and deductions, will have...

increased by 12% or more (bin 1)	_____	percent chance
increased by 8% to 12% (bin 2)	_____	percent chance
increased by 4% to 8% (bin 3)	_____	percent chance
increased by 2% to 4% (bin 4)	_____	percent chance
increased by 0% to 2% (bin 5)	_____	percent chance
decreased by 0% to 2% (bin 6)	_____	percent chance
decreased by 2% to 4% (bin 7)	_____	percent chance
decreased by 4% to 8% (bin 8)	_____	percent chance
decreased by 8% to 12% (bin 9)	_____	percent chance
decreased by 12% or more (bin 10)	_____	percent chance

Data: NY Fed Survey of Consumer Expectations

SCE. Monthly rotating panel, NY Fed, 2013Q2–present. Up to 12 consecutive months per respondent.

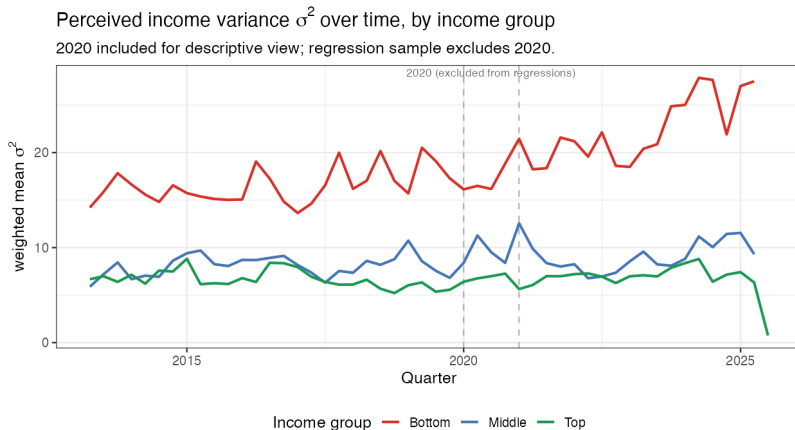
Key variable: density forecast. The survey asks for the probability distribution over next-year earnings growth on the respondent's *current main job*. Respondents place mass $p_{i,t,k}$ across 10 pre-set bins. From this:

$$\mu_{i,t}^{\text{subj}} = \sum_k p_{i,t,k} m_k, \quad \sigma_{i,t}^2 = \sum_k p_{i,t,k} (m_k - \mu_{i,t}^{\text{subj}})^2$$

Sample filters. Working-age (18–64), employed. *We only keep respondents who stay in the same job throughout their spell*, so the forecast question stays about that same job. Consecutive person spells containing ≥ 4 quarters retained. Calendar 2020 excluded due to a sharp drop in survey respondents during COVID.

Final sample. $\sim 26,000$ person-quarter obs (median tenure 5 months). Three income groups, dollar-denominated: Bottom ($< \$50k$), Middle ($\$50\text{--}100k$), Top ($\geq \$100k$).

Perceived risk over time

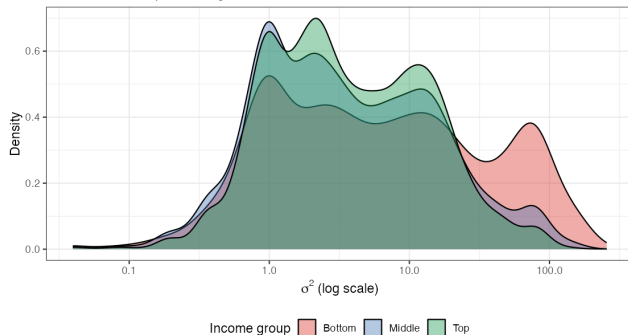


Weighted mean σ^2 by income group, 2013Q2–2025Q3. 2020 shown for descriptive view but excluded from the regression sample. The income-graded ranking is stable throughout; gap does not close.

Perceived risk is income-graded in levels

Distribution of σ^2 by income group

Pooled across all quarters; log scale x-axis



Pooled density of σ^2 by income group.

Group	Mean σ^2	N
Bottom	25.13	3,709
Middle	11.99	6,375
Top	9.87	7,455

Mean σ^2 and person-quarter count by income group.

- Bottom shifted right with a heavier right tail; 2.55 $\times$ ratio Bottom-to-Top in mean perceived variance.
- **Not new.** Levels taken as given. Rest of the paper asks whether elevated risk states are *persistent*, *respond to shocks*, and whether this differs by income.

Exploratory Regressions

Persistence is income-graded; but only in beliefs

Question: is the income-persistence gradient just because objective income is noisier at the bottom?

SCE (perceived): regress $\sigma_{i,t}^2$ on its lag, SEs clustered respondent $\times$ quarter. **PSID (realized):** ARCH(1) on squared wage innovations 1970–2021, clustered household $\times$ year (Meghir & Pistaferri 2004). Both sides residualised for demographics, time effects absorbed.

	<i>SCE: perceived</i> $\hat{\lambda}$	<i>PSID: realized</i> ARCH $\hat{\alpha}_1$
<\$50k (Bottom)	0.926***	0.174***
\$50–100k (Middle)	0.880***	0.186***
>\$100k (Top)	0.857***	0.163***
Bottom $\neq$ Top	$p < 0.001$	$p = 0.768$

$\Rightarrow$ Sharp gradient in perceived risk. PSID realized income shows no such gradient. The persistence gradient is a **belief distortion**, not a feature of the objective income process.

Estimating within-person persistence

We're after the *dynamics* of perceived risk, so we need within-person variation.

- **Pooled OLS by group.** What looks like persistence could be driven by differences *between* people rather than *within*. Most σ^2 variation is idiosyncratic across respondents, see the observed factor decomposition:
Patterson (2023) style .
- **Direct route: person FE.** Nickell bias (1981) compresses $\hat{\lambda}$ toward zero on short panels (up to 12 months).
- **Solution: system GMM (Blundell & Bond 1998).** First-difference removes α_i ; lagged levels instrument lagged differences.
- **Precedent.** Yueran Ma (2020): AR(1), very similar estimation context, high persistence on a short panel of managers' forecast errors.

Within-person persistence (GMM)

System GMM (Blundell & Bond 1998), quarterly, two-step, two-way clustered SEs (user $\times$ quarter), collapsed instruments. full spec 33-spec robustness

$$\sigma_{i,t}^2 = \lambda_g \sigma_{i,t-1}^2 + \alpha_i + \varepsilon_{i,t}$$

Instruments: diff. eq. $\{\sigma_{i,t-2}^2, \sigma_{i,t-3}^2\}$; level eq. BB moment $\Delta\sigma_{i,t-1}^2$.

Group	$\hat{\lambda}_g$	SE	Hansen J p
<\$50k (Bottom)	+ 0.575***	0.133	0.50
\$50–100k (Middle)	+0.356**	0.151	0.049
>\$100k (Top)	+0.207***	0.016	0.66

⇒ **Bottom carries last quarter's perceived risk forward stronger than Top.** The income gradient survives instrumenting and within-person identification. Hansen J test of overidentifying restrictions not rejected at Bottom and Top; borderline at Middle.

The Shock

The shock measure: income forecast errors

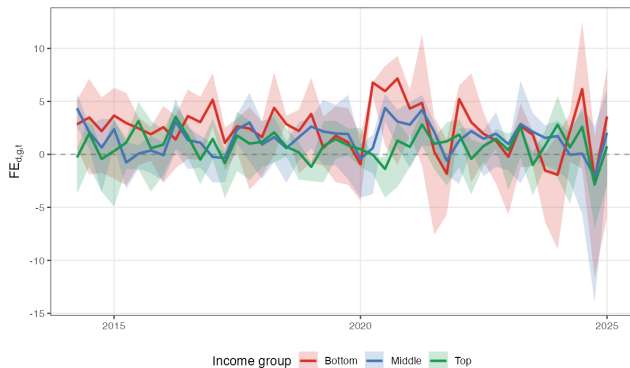
Challenge. Individual income surprises are not observed in the SCE.

Solution. Following Guerreiro (JMP), proxy the shock by allocating it to each respondent based on their Census Division (9 divisions) $\times$ income group $\times$ month cell. Narrow enough to capture group-specific shocks, broad enough for sample size. The forecast error is the gap:

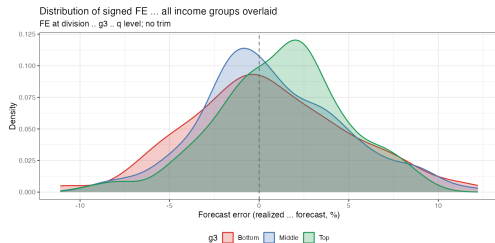
$$FE_{d,g,t} \equiv \underbrace{\bar{g}_{d,g,t}^{\text{CPS}}}_{\text{CPS 12-month earnings growth}} - \underbrace{\bar{\mu}_{d,g,t-12}^{\text{SCE}}}_{\text{SCE expected growth (lagged 12m)}}$$

- **Construction.** CPS realization: 12-month earnings change, same person at same employer. SCE forecast: average of individual density forecasts within each cell. Both sides residualized on {education, age, age², sex, race}.
- **No year fixed effect.** Year FE would absorb macro variation. But macro variation is exactly the surprise we want to measure, so years stay in on both sides.
- **Final FE panel.** 526 division-group-quarter cells (2020 excluded). By income group: Bottom 110, Middle 205, Top 211.

Distribution of forecast errors by income group



Cross-division spread of $FE_{d,g,t}$ over time. Weighted mean (line) and IQR (25–75 band) across divisions.



Pooled density of $FE_{d,g,t}$ by income group. Bottom skews positive; Top skews negative.

Bottom updates more on bad news than good news

System GMM, 2-step weighted, two-way clustered (respondent $\times$ qdate). $\hat{\lambda}_g$ within-person; $\hat{c}_g^\pm$ on cross-cell variation (division $\times$ g3 $\times$ qdate). $FE^+ = \max(FE, 0)$, $FE^- = -\min(FE, 0) \geq 0$. GMM specification Pooled $|FE|$ robustness result

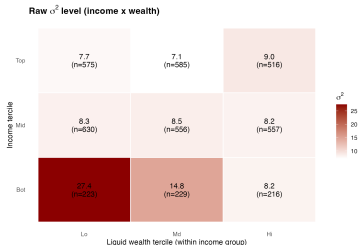
$$\sigma_{i,t}^2 = \lambda_g \sigma_{i,t-1}^2 + c_g^+ FE_{d,g,t}^+ + c_g^- FE_{d,g,t}^- + \alpha_i + \varepsilon_{i,t}$$

Group	$\hat{\lambda}_g$	$\hat{c}_g^+$	$\hat{c}_g^-$
<\$50k (Bottom)	0.655***	+0.496***	+1.203***
\$50–100k (Middle)	0.507**	−0.043	+0.013
>\$100k (Top)	0.148***	−0.270	−0.316*
<i>Wald p (B = T)</i>	0.003	0.003	0.001

⇒ **Shock channel concentrated at the Bottom.** Bot loads on both signs; point estimate $\hat{c}_B^- / \hat{c}_B^+ \approx 2.4$. Mid and Top insignificant on both shocks.

Wealth matters too: the income gradient lives at low wealth

Raw σ^2 level (income $\times$ wealth)



Pooled OLS persistence $\hat{\lambda}_g^{\text{OLS}}$ (3 income $\times$ 2 wealth)

group	Lo wealth	Hi wealth
Bot	0.906*** (0.032)	0.733*** (0.086)
Mid	0.558*** (0.061)	0.854*** (0.064)
Top	0.537*** (0.084)	0.785*** (0.033)
$p(\lambda_B = \lambda_T)$	< 0.001 0.62	

Wealth split = within-income median liquid savings.

⇒ **(left) σ^2 level:** Bottom highest, with a sharp within-Bottom gradient over wealth

⇒ **(right) Persistence:** the income gradient lives at low wealth (Bot $\gg$ Top); at high wealth, all three income groups persist at roughly the same rate.

⇒ **Useful moment for model calibration.** Both patterns point to the same mechanism: low-wealth (near-constraint) households drive the high-risk, high-persistence corner.

Caveat. SCE $\sigma^2 \rightarrow$ HHF merge drops $\sim 60\%$ of users. Pooled OLS used because GMM is underpowered per cell.

Model

Model: Aiyagari¹ with $\tilde{\sigma}^2$ as a Bellman state

Bellman.

$$V(a, y, \tilde{\sigma}^2) = \max_c u(c) + \beta \mathbb{E}_{\tilde{\sigma}^2}[V(a', y', \tilde{\sigma}^{2'}) | y], \quad a' = (1 + r)a + we^y - c, \quad a' \geq 0$$

CRRA $\gamma = 2$, $r = 0.03/4 = 0.0075$, quarterly stride. Single β across terciles.

Three state variables.

- a – assets (buffer-stock savings, borrowing constraint $a \geq 0$).
- y – current income (AR(1), discretised on 7-state Rouwenhorst grid).
- $\tilde{\sigma}^2$ – agent's *perceived* variance of 12-month-ahead income growth.

$\Rightarrow$ Standard buffer-stock with one extra state: time-varying perceived risk $\tilde{\sigma}^2$. Wider $\tilde{\sigma}^2 \Rightarrow$ stronger precautionary motive. Evolution of $\tilde{\sigma}^2$ next slide.

¹I know it's not really an Aiyagari yet, but rather a Bewley-Huggett at fixed r ; this will hopefully be the next step...

Diagnostic LOM for perceived $\tilde{\sigma}^2$

$$\tilde{\sigma}_{i,t}^2 = \underbrace{\phi(a, y) \lambda_0 \tilde{\sigma}_{i,t-1}^2}_{\text{persistence}} + \underbrace{\phi(a, y) \kappa_0 [\tilde{\varepsilon}_{i,t}^+ + \omega \tilde{\varepsilon}_{i,t}^-]}_{\text{response (asymmetric in } \omega)} + \underbrace{(1 - \phi(a, y)) \bar{\sigma}^2}_{\text{drift to common prior}}$$

$\phi(a, y) \in [\phi_{\min}, \phi_{\max}]$: state-dependent weight on the LOM dynamics (call it “attention”). Form and motivation in next slides.

Three components.

- *Persistence*: $\tilde{\sigma}^2$ carries forward with effective rate $\phi \cdot \lambda_0$.
- *Asymmetric response*: $\tilde{\sigma}^2$ jumps with current shocks at rate $\phi \cdot \kappa_0$; $\omega \geq 1$ amplifies negatives.
- *Drift*: weight $(1 - \phi)$ pulls $\tilde{\sigma}^2$ toward the common prior $\bar{\sigma}^2$.

$\Rightarrow \phi = 0$ **recovers RE**: $\tilde{\sigma}^2$ stays at $\bar{\sigma}^2$ forever, agent solves standard Aiyagari with constant perceived variance. The model nests rational expectations as a limit case.

Microfoundation. Reduced form: state-dependent diagnostic intensity à la Maxted (2023 *JF*); nests constant- θ diagnostic expectations (Bordalo, Gennaioli & Shleifer 2018 *JF*) as the ϕ equal across agents.

Why state-dependent attention $\phi(a, y)$?

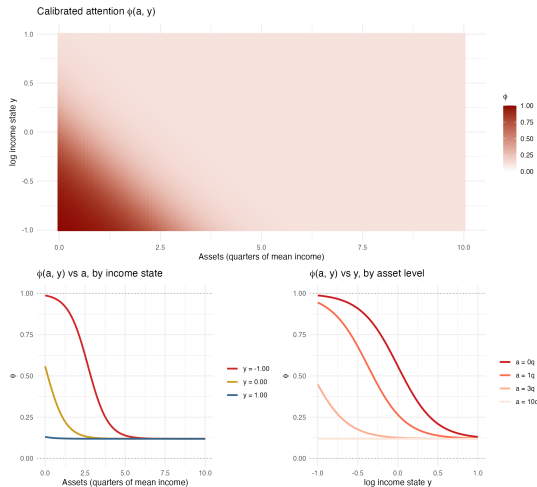
Strategy. We pin down what ϕ must look like from the data; microfoundation postponed for now.

- ϕ **must vary across income.** A common ϕ predicts identical persistence and asymmetric response for every agent. SCE shows clear heterogeneity in $\hat{\lambda}$ across income terciles, so I let ϕ vary.
- ϕ **must also depend on wealth.** Within Bot tercile, persistence declines steeply with wealth ($0.92 \rightarrow 0.84 \rightarrow 0.72$, see [past slide](#)). Bot agents with savings behave a bit more like Top.
- **Stakes per shock.** Constrained agents feel shocks 1-for-1 in consumption; buffered agents absorb them from savings. Attention pays off in the first case, barely in the second. Both y and a determine the stake.

$\Rightarrow \phi(a, y)$: one function, no income- or wealth-indexed parameters. I present the functional form next.

Antecedents. Stake-weighted attention: Gabaix (2014 *QJE*). State-dependent diagnostic intensity: Maxted (2023 *JF*). Asymmetric attention to good vs bad news: Kohlhas & Walther (2021 *AER*). Foundational DE: Bordalo, Gennaioli & Shleifer (2018 *JF*). Rational inattention: Maćkowiak & Wiederholt (2009, 2015).

Functional form: standard LSTAR architecture



Logistic smooth-transition (LSTAR).

$$\phi(a, y) = \phi_{\min} + (\phi_{\max} - \phi_{\min}) \cdot \mathcal{L}(-(k_a a + k_y y))$$

$\mathcal{L}(z) = (1 + e^{-z})^{-1}$. Bounded, smooth and monotone.

Precedent. *Auerbach & Gorodnichenko (2012)* use an LSTAR with a single state variable (cyclical slack) to switch fiscal multipliers smoothly between recession and expansion regimes. We adopt the same functional form, with two state variables (a, y) as the switching arguments and attention ϕ as the regime-varying object.

Also: Tenreyro & Thwaites (2016 *AEJ:Macro*) for state-dependent monetary policy; Ramey & Zubairy (2018 *JPE*) for slack-state fiscal multipliers.

Calibration: 10 moments $\rightarrow$ 8 free parameters

Simulate panel ($N=5000$, $T=160q$) $\rightarrow$ run identical GMM-asym as on SCE $\rightarrow$ match the same coefficients.
Five sign restrictions on derivatives confirmed pre-calibration: [appendix](#).

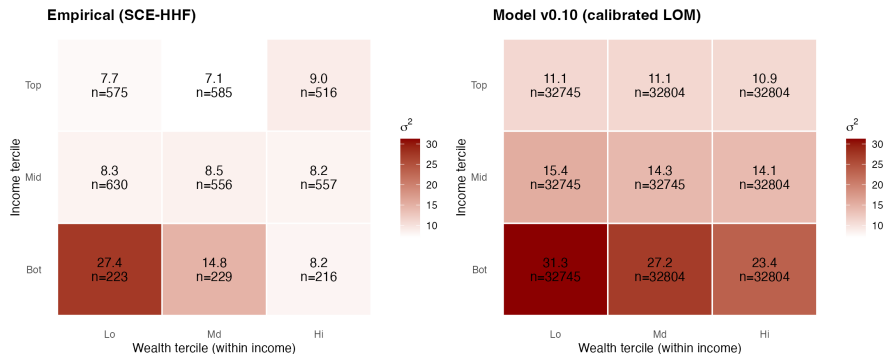
Fixed $\bar{\sigma}^2=10$, $\gamma=2$, $r=0.03/4$, $\rho_y^{\text{ann}}=0.97$, $\sigma_{\varepsilon}^{\text{obj,ann}}=0.16$

Calibrated $\lambda_0=0.77$, $\kappa_0=5.62$, $\omega=0.84$, $\phi_{\max}=0.999$, $\phi_{\min}=0.12$, $k_a=7.82$, $k_y=4.33$, $\beta=0.994$

moment	emp	sim	moment	emp	sim
$\hat{\lambda}_{\text{Bot}}$	0.66	0.61	$\tilde{\sigma}_{\text{Bot}}^2/\tilde{\sigma}_{\text{Top}}^2$	2.55	2.46
$\hat{\lambda}_{\text{Mid}}$	0.51	0.35	$\tilde{\sigma}_{\text{Mid}}^2/\tilde{\sigma}_{\text{Top}}^2$	1.21	1.32
$\hat{\lambda}_{\text{Top}}$	0.15	0.16	$w\text{-median}_{\text{Bot}}(q)$	0.23	0.22
$\hat{c}_{\text{Bot}}^+$	0.50	0.45	$w\text{-median}_{\text{Mid}}(q)$	0.46	0.66
$\hat{c}_{\text{Bot}}^-$	1.20	1.09	$w\text{-median}_{\text{Top}}(q)$	2.31	1.63

Untargeted check: σ^2 level on the (*income*, *wealth*) grid

Cell means of σ^2 on a 3×3 grid: income tercile × within-income liquid-wealth tercile.

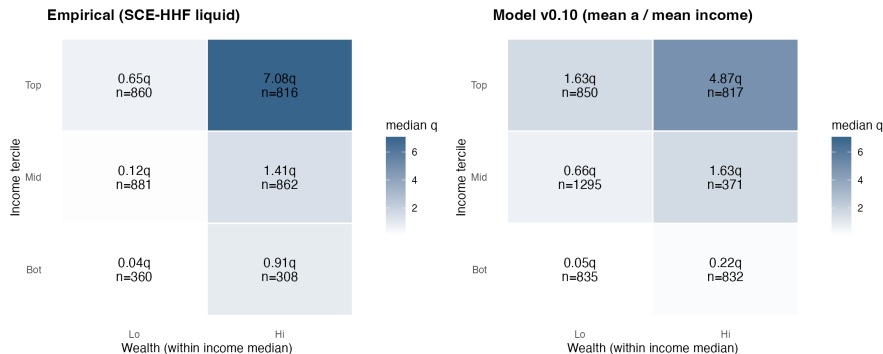


⇒ **Cross-tercile ordering matches.** Bot > Mid > Top in both panels, plus a clear within-Bot wealth gradient (decreasing in both).

⇒ **Magnitude mismatch:** Bot drops $\sim 3.4\times$ across wealth in the data vs only $\sim 1.3\times$ in the model.

Untargeted check: liquid wealth on the (*income*, *wealth*) grid

Median liquid wealth in quarters of cross-sectional mean quarterly income.

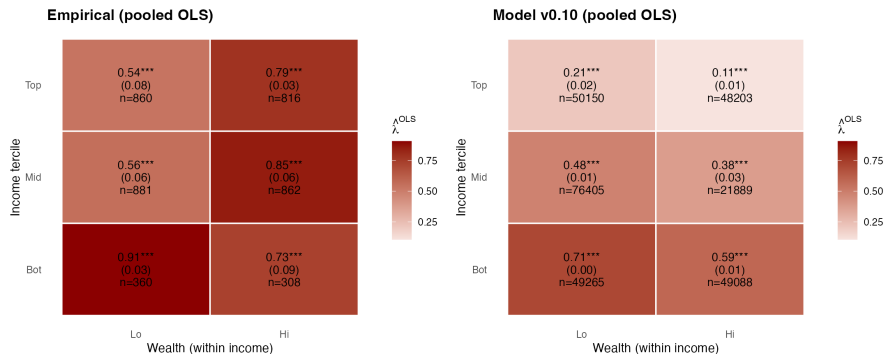


⇒ **Ordering matches.** Bot < Mid < Top and Lo < Hi in both panels.

⇒ **Right tail compressed in the model.** Single- β Aiyagari can't generate the high-wealth dispersion seen in empirics. A two-asset or β -heterogeneity extension could help.

Untargeted check: persistence on the (*income*, *wealth*) grid

Pooled OLS $\hat{\lambda}$ within each (income tercile $\times$ Lo/Hi wealth) cell. Wealth split at within-income median liquid savings.

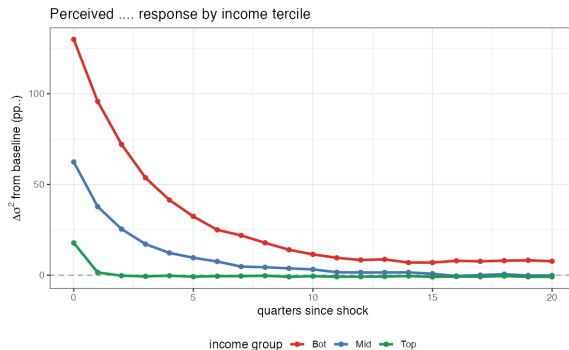


⇒ **Low wealth: matches.** Persistence falls across income terciles in both panels (the high- ϕ corner).

⇒ **High wealth: doesn't match.** Empirical $\hat{\lambda}$ stays elevated, while the model predicts a stronger buffer-dampening than what we see in the data. The high-wealth buffer doesn't kill persistence as much to match empirics.

Action channel: perceived σ^2 IRF by income tercile

Shock: -1SD income realization. **Channel.** Income shock $\rightarrow$ LOM updates σ^2 , scaled by attention $\phi(a, y)$.
Bot at high ϕ (near constraint) amplifies; Top at low ϕ (well-buffered) tunes out.

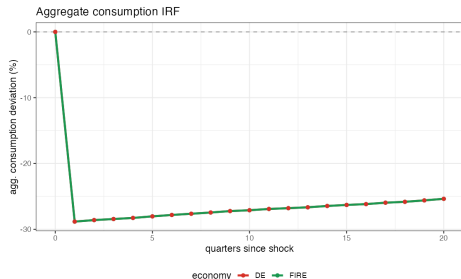


$\Rightarrow$ **Bottom amplifies and stays elevated** for many quarters; **Top mean-reverts** within a few quarters.

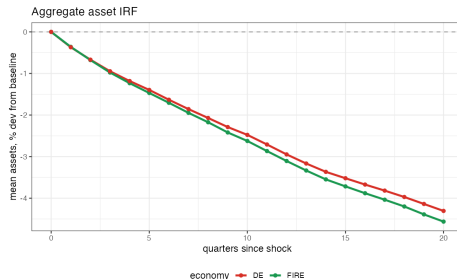
Aggregate response: consumption vs assets

Channel. Same shock $\rightarrow$ Higher $\sigma^2 \rightarrow$ wider perceived risk $\rightarrow$ stronger precautionary motive $\rightarrow$ less consumption, more saving. **FIRE** benchmark: $\sigma^2 \equiv \bar{\sigma}^2$, no LOM.

Consumption



Assets



- $\Rightarrow$ **Consumption:** dominated by the direct shock; DE–FIRE wedge only ± 0.02 pp.
- $\Rightarrow$ **Saving margin shows the friction.** Bot wants to save more but the constraint binds. Mid is the sweet spot (buffer + σ^2 spike): +0.78 pp vs FIRE. Top: buffer but no σ^2 signal.
- $\Rightarrow$ **Aggregate result:** +0.26 pp at $q=20$, +1.23% more wealth in steady state.

From PE to GE

My current thoughts: In PE the aggregate precautionary effect is small. +1.23% more wealth in steady state, consumption gap ± 0.02 pp. I have a few PE knobs left to play with like risk aversion γ , other preference choices, finer income grid. Maybe the bigger story lies in GE..?

Why GE could matter. In PE r is fixed; the extra saving under DE just sits in the buffer. In GE the higher K has to clear against capital demand, so r adjusts, which feeds back into saving and labor supply. After a shock, the transient +0.26 pp asset wedge could also push r dynamically. The aggregate response could amplify or dampen.

Next steps.

- **First and foremost.** Microfound $\phi(a, y)$ from a costly-attention / stake-weighted choice problem, then derive an analytical wedge directly rather than imposing a reduced-form LOM.
- **Preferences.** Play with CRRA γ (currently 2), or try different preference families to see how much the precautionary motive can be amplified within PE (how prudence can help).
- **Close the loop.** Feed the calibrated $(\phi, \lambda_0, \kappa_0, \omega)$ into a closed economy, let r adjust, see what it does to aggregate consumption, labor, and the dynamics of r itself.

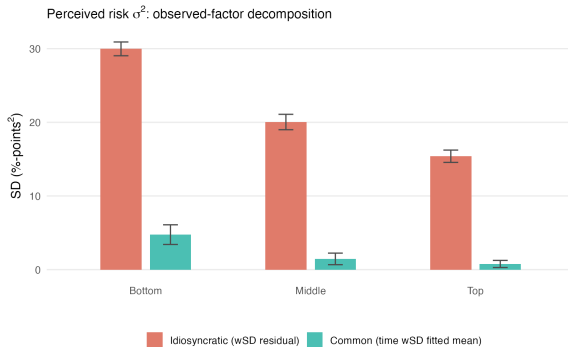
- 1 **One mechanism delivers all three SCE facts.** A standard buffer-stock with state-dependent attention $\phi(a, y)$ and sign asymmetry ω generates the cross-tercile gradient in level, persistence, and asymmetric updating from common primitives. The heterogeneity emerges as a *composition effect* of the joint (a, y) distribution under one attention function.
- 2 **Cross-group differences are equilibrium artifacts of the wealth distribution.** Bot agents stack at high ϕ ; Top at low ϕ . Heterogeneity is not assumed; it emerges from where each tercile sits in (a, y) space.
- 3 **Aggregate IRFs depend on the wealth distribution.** Perceived risk stays elevated for longer at the bottom, dragging consumption recovery. Redistribution shifts perceived risk and therefore the aggregate response.

Appendix

Appendix: perceived risk is overwhelmingly idiosyncratic

Setup. A la Patterson (2023). F_t = cross-sectional weighted mean of $\sigma_{i,t}^2$:

$$\sigma_{i,t}^2 = \alpha_g + \beta_g F_t + e_{i,t}$$



- ⇒ **Within-person dynamics dominate.** Idiosyncratic variation is first-order everywhere.
- ⇒ **Aggregate exposure is income-graded.** Common component much larger at Bottom.

Appendix: System GMM specification

Model. $\sigma_{i,t}^2 = \alpha_i + \lambda_g \sigma_{i,t-1}^2 + c_g^+ FE_{d,g,t}^+ + c_g^- FE_{d,g,t}^- + \varepsilon_{i,t}$

Estimator. System GMM (Blundell & Bond 1998), two-step weighted, transform = Id. Hand-rolled to handle two-step weighting with depth-IV; validated against Stata. Standard panel-GMM libraries mishandle this combination.

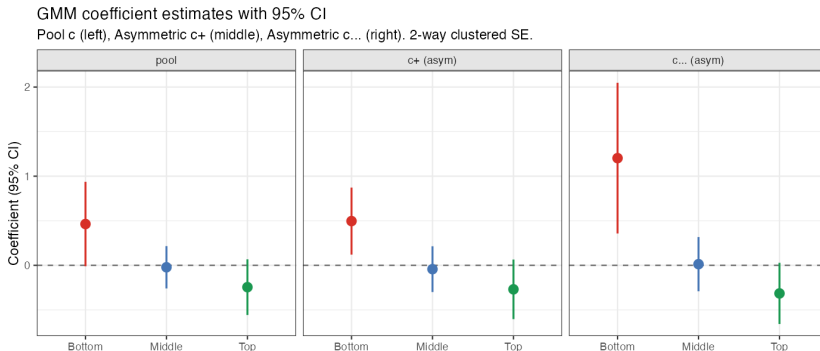
Instruments.

- Difference eq.: $\sigma_{i,t-2}^2, \sigma_{i,t-3}^2; |FE|_{d,g,t}, |FE|_{d,g,t-1}$ (predetermined).
- Level eq.: $\Delta \sigma_{i,t-1}^2, \Delta |FE|_{d,g,t}$.

Cluster. Two-way respondent $\times$ quarter. Robustness across cluster choices: 33-spec sweep.

Validity. Hansen J , AR(2) on differenced residuals, Sargan: all not rejected under T5-asym (full sample).

Appendix: 33-spec robustness sweep



$\hat{\lambda}_B, \hat{\lambda}_M, \hat{\lambda}_T$ across 33 perturbations of the baseline: transform, IV depth, cluster, residualization, sample window.

⇒ **Headline preserved.** All point estimates within ± 0.07 ; income-group ordering holds in every spec.

⇒ **Caveats.** (i) sweep is on $\hat{\lambda}$ only; $\hat{c}^\pm$ stability still TBD. (ii) Mid Hansen J $p = 0.026$ at headline: borderline rejection, point estimate survives. (iii) Cell-FE shock and σ^2 -lag share the panel (reflection concern); leave-one-out cell means is the natural fix.

Appendix: pooled $|FE|$ specification

Spec. Same GMM, same instruments, single pooled shock loading $c_g |FE|_{d,g,t}$ in place of $c_g^\pm$.

Group	$\hat{\lambda}_g$	$\hat{c}_g$ (pooled $ FE $)
<\$50k (Bottom)	0.557***	+ 0.463**
\$50–100k (Middle)	0.372*	−0.022
>\$100k (Top)	0.178***	−0.245
<i>Wald p (B = T)</i>	0.062	0.014

⇒ **Same tercile conclusion.** Shock channel concentrated at Bottom.

⇒ **Why main slide uses asymmetric spec.** Pooled c_g is a sign-averaged version of $c_g^\pm$; it hides the $\sim 2.4\times$ asymmetric loading inside Bottom's $\hat{c}$.

Appendix: pre-calibration sign restrictions

Setup. Differentiating the LOM gives five sign predictions from $\phi, \lambda_0, \kappa_0, \omega > 0$ and ϕ decreasing in (a, y) , before any calibration.

#	Predicted derivative	Sign	Data
1	$\partial \tilde{\sigma}_{,t}^2 / \partial \tilde{\sigma}_{,t-1}^2 = \phi \lambda_0$	+	✓ Confirmed ($p < .001$, all terciles)
2	$\partial \tilde{\sigma}_{,t}^2 / \partial \tilde{\varepsilon}^- = \phi \kappa_0 \omega$	+	✓ Confirmed (Bot $\hat{c}^- = 1.20$)
3	$\mathbb{E}[\tilde{\sigma}^2 g]$ decreasing in income rank g	−	✓ Confirmed (level gradient 25/12/10)
4	$\partial^2 \tilde{\sigma}_{,t}^2 / \partial \tilde{\sigma}_{,t-1}^2 \partial g$	−	✓ Confirmed ($\hat{\lambda}_B = 0.66 \gg \hat{\lambda}_T = 0.15$)
5	$\partial^2 \tilde{\sigma}_{,t}^2 / \partial \tilde{\varepsilon}^- \partial g$	−	✓ Confirmed (Bot $\hat{c}^-$ dominates)

⇒ **Five signs from one $\phi(a, y)$ plus one ω .** No tercile-conditional primitives. Calibration tests whether magnitudes also line up.

Appendix: The model action channel, but compressed

Bellman. $V(a, y, \sigma^2)$, single $\beta = 0.994$, cash-on-hand $= R a + \exp(y)$.

Grids. Asset $a \in [0, 60]$, $n_a = 60$, curve = 2. Income: Rouwenhorst HSV, $\rho_q = 0.992$, $\sigma_\varepsilon^{\text{obj}, q} = 0.0809$, $n_y = 7$.

Perceived $P_y(\sigma^2)$. Tauchen on HSV y -grid with $\sigma_\varepsilon^{\text{perc}}(\sigma^2) = \sqrt{\sigma^2}/100$. (σ^2 is the agent's perceived 1-quarter innovation variance in pp^2 .)

Caveat. Differs from the LOM-calibration mapping $\sqrt{\tilde{\sigma}^2/(100^2 \cdot \text{sce_var_mult})} \approx 0.016$, which would give a near-identity P_y and a near-zero action channel.

σ^2 **LOM (same as calibration).** $\sigma^{2'} = \phi(a, y) [\lambda_0 \sigma^2 + \kappa_0(\tilde{\varepsilon}^+ + \omega \tilde{\varepsilon}^-)] + (1 - \phi(a, y)) \tilde{\sigma}^2$.

FIRE. $\sigma^2 \equiv \tilde{\sigma}^2 = 10$ for all agents, no LOM. Same income grid, same β , same γ .

Simulation. $N = 5000$, $T = 160q$ (burn 100, keep 60). Tercile labels from long-run mean $\exp(y)$ under the *objective* P_y (independent of β, σ^2).

Shock. $t = T_{\text{burn}} + 1$: each agent's income index drops 1 grid step ($\approx -40\%$). FIRE and DE share the path.

Calibrated (8). $\lambda_0 = 0.77$, $\kappa_0 = 5.62$, $\omega = 0.84$, $\phi_{\max} = 0.999$, $\phi_{\min} = 0.12$, $k_a = 7.82$, $k_y = 4.33$, $\beta = 0.994$. $\tilde{\sigma}^2 = 10$ fixed.